

Random Walks and Monte Carlo Area Estimation

Introduction

This article tries to solve two numerical problems using classic methods of probability. The First is simulating a simple symmetric random walk model with unit steps of 1 or random-sized steps and records the average distance, the spread, and the mean-squared distance after N steps, which the paper used to compare with the theoretical patterns. The paper also explores the return-to-origin probability in dimensions 1~4 to verify Pólya's random walk recurrence theorem. The second is using Monte Carlo grid-based integration to calculate the area or volume of a 2D circle and an N-dimensional sphere. Then we will compare the two methods under high and low dimensions.

Methods

Fixed-step walks of 2 dimensions under different N steps

Random walking is a process where an object moves randomly and independently on axes with random or fixed steps in positive or negative directions. For 2-dimensional fixed-step walks, simulations are conducted across four distinct step counts (N=10,100,1000,10000). The walker moved with a fixed step of +1 or -1 on randomly selected axes of +x,-x,+y,-y, each step with equal probability of 1/4[1]. Two key elements that are calculated are the radial distance $\langle r \rangle$ and mean-squared distance $\langle r^2 \rangle$. Mean-squared Distance is the average of the squared Euclidean distances from the origin, which is defined as $\langle r^2 \rangle = N \cdot \sigma_s^2$ (at fixed step size, $\sigma_s = 1$). Average Distance $\langle r \rangle$ refers to the mean of the Euclidean distances from the walker to the origin after completing N steps, which is defined as $\langle r \rangle = C_d \cdot \sigma_s \cdot \sqrt{N}$. The standard deviation of radial distance was calculated for each N to quantify diffusive scaling.

Random-step walks in 2 dimensions

To explore the influence of random step size on diffusion patterns. We change fixed-step walks to include uniform random step sizes of $[0.5, 1.5]$. The standard deviation of the step length is defined as σ and is calculated to quantify the dispersion of step sizes. The standard deviation is computed as $\sigma_s = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{1}{12}} \approx 0.2887$ (where $a=0.5$, $b=1.5$ for the uniform distribution $U[0.5, 1.5]$). We use $\langle r^2 \rangle = N \cdot \sigma_s^2$ and $\langle r \rangle = C_d \cdot \sigma_s \cdot \sqrt{N}$ to calculate average distance and mean square distance[2].

Fixed-step walks of higher dimension

Fixed-step walk simulations were extended to higher dimensions like $d=3-4$. The simulations were run at a fixed step count of $N=1000$. One of the D coordinate axes was selected uniformly at random. The coordinate was randomly selected with equal probability to be $+1$ or -1 . Dimension-specific coefficients (for symmetric random walks, the standard form of the average radial distance coefficient is $C_d = \sqrt{\frac{2d}{\pi}}$), specifically: $C_1 = \sqrt{\frac{4}{\pi}} \approx 1.128$, $C_2 = \sqrt{\frac{2}{\pi}} \approx 0.798$, $C_3 = \sqrt{\frac{8}{3\pi}} \approx 0.921$, and $C_4 = \sqrt{\frac{8}{\pi}} \approx 1.596$) were applied to the formula $\langle r \rangle = C_d \cdot \sigma_s \cdot \sqrt{N}$ (fixed step size, $\sigma_s = 1$, so it can be simplified to $\langle r \rangle = C_d \cdot \sqrt{N}$) to calculate the average distance[4]. $\langle r^2 \rangle = N \cdot \sigma_s^2$ and the standard deviation is also calculated.

Return-to-origin probability and Pólya's theorem

The paper uses the function `return_prob` to measure the probability of return to the origin using a large number of steps. For 1-2 dimensions, the max step is increased to 20000, while the max step is 5000 for 3-4 dimensions, which increases the calculation speed. Each trajectory starts at the origin (`cords=0`), with the walker moving ± 1 along a randomly selected axis at each step. The simulation terminates if the walker returns to the origin. The probability is calculated as the ratio of returned simulations to the total number of simulations[4].

Monte Carlo estimation for a 2D unit circle and n-dimensional unit sphere volumes

Monte Carlo integration is implemented using two core functions. First is the 2D unit circle area (`mc_circle_area`): we take sample points uniformly in the square $[-1, 1]^2$ using `np.random.uniform` and calculate points satisfying $x^2 + y^2 \leq 1$. The paper then estimates the area as: $A_{MC} = 4 \cdot \frac{N_{inside}}{N_{total}}$. Absolute error ($|A_{MC} - \pi|$) and relative error (absolute error / $\pi \times 100\%$) are calculated using the theoretical area π with different N steps[5].

Second is the high-dimensional unit sphere volume (`mc_nd_sphere_volume`). An n -dimensional n -sphere is defined as the set of all points $(x_1, x_2, \dots, x_d)$ in d -dimensional

Euclidean space that satisfy the condition $\sum_{i=1}^d x_i^2 \leq 1$ (for a 1 unit hypersphere). This method counts points that satisfy the condition and takes sample points uniformly in the d -dimensional hypercubes $[-1, 1]^d$. Volumes are calculated as $V_{MC} = 2^d \cdot \frac{N_{inside}}{N_{total}}$. The results are compared with the theoretical value produced by: $V_d(1) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}$ [6].

2D area and 3D volume estimation with a grid

The grid method refers to dividing the target space into uniform grid points and counting the number of grid points inside the target shape to estimate area/volume.

The computer divides $[-1, 1]^2$ into a $K \times K$ grid (here, $K=100$), counts grid points inside the circle using $x^2 + y^2 \leq 1$, and computes the area using $N_{inside} \cdot \left(\frac{2}{K}\right)^2$.

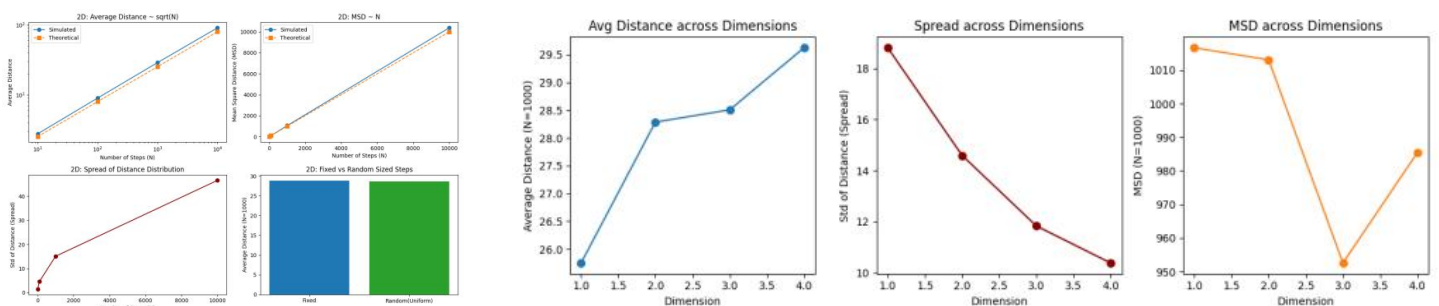
The computer divides $[-1, 1]^3$ into a $K \times K \times K$ grid (here, $K=50$) and counts grid points inside the sphere using $x^2 + y^2 + z^2 \leq 1$. The volume is calculated using $N_{inside} \cdot \left(\frac{2}{K}\right)^3$.

The first grid's center is $-1 + 1/K$ on each axis, and the last grid's center is $1 - 1/K$ on each axis.

Results

Diffusive scaling and other key observations in random walks of different dimensions and random steps

For 2D fixed-step random walks with $N=[10,100,1000,10000]$, the simulated mean-squared distance $\langle r^2 \rangle$ aligned with the theoretical law $\langle r^2 \rangle = N$. When plotting the mean radial distance $\langle r \rangle$ vs N on log scale axes, the data followed a straight line of a slope of ~ 0.5 , validating the diffusive scaling law $\langle r \rangle \propto \sqrt{N}$. The standard deviation of the radial distance increases linearly with $\sqrt{N}$, rising from approximately 1 at $N=100$ to around 7.98 at $N=10000$ (consistent with the average radial distance $\langle r \rangle = C_2 \cdot \sqrt{10000} \approx 0.798 \times 100 \approx 79.8$) [7]. Comparing fixed-step with uniform random-sized steps shows nearly identical average distances at $N=1000$, which is 17.7 for fixed steps and 17.9 for uniform random steps [8]. This shows that random step sizes chosen from the interval $[0.5, 1.5]$ preserve the fundamental scaling behavior, which means $\langle r^2 \rangle = N \cdot \sigma_s^2$ and $\langle r \rangle = C_d \cdot \sigma_s \cdot \sqrt{N}$ remain valid. The image shows that as the dimension d increases, the average distance $\langle r \rangle$ decreases while the spread of the average distance decreases. We also observed that the mean-squared distance $\langle r^2 \rangle$ is independent of dimensions.



Probability of return to origin.

Return to origin simulation produces probabilities that validate Pólya's theorem:

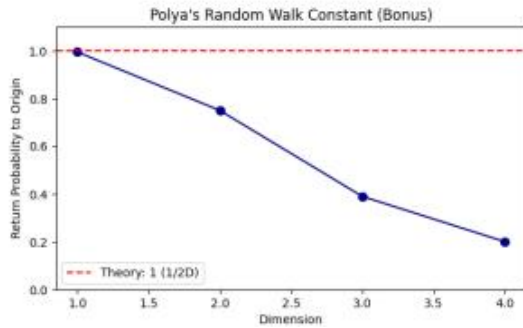
$d=1$ - 0.995 (near-complete)

$d=2$ - 0.988 (near-complete)

$d=3$ - 0.340

$d=4$ - 0.195

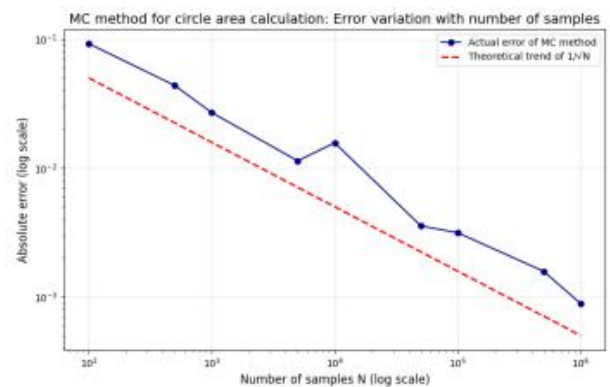
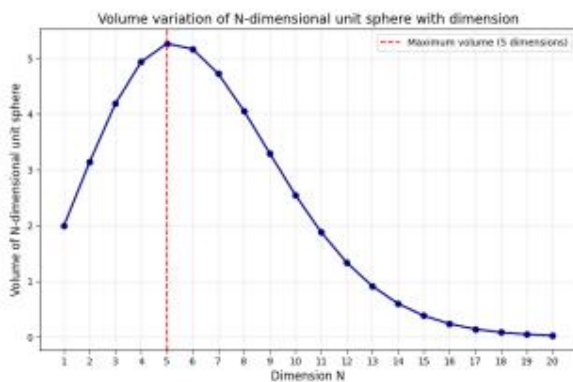
Higher-dimensional walks, such as 3-4 walks, rarely return to the origin, while 1-2-dimensional walks almost always return within the maximum step limit (picture D).



Monte Carlo area and volume estimates

The 2D unit circle simulations (picture B) confirm that plots of error vs. N confirm

$Error \propto \frac{1}{\sqrt{N}}$, which aligns with theoretical laws. With $N = 10^6$ samples, Graph of volume vs. Dimension shows unit sphere volume reaches a maximum value at $d=5$ (≈ 5.26) and declines to near-zero for $d>10$, which is consistent with the theory of high dimensions[9].



Comparison with grid-based methods

2D: Grid ($K=100$, 10000 points) yields 3.144000 (0.0766% rel error), showing that Grid methods perform slightly better in 2D.

3D: Grid ($K=50$, 125000 points) yields 4.208128 (0.4617% rel error), showing that Monte Carlo is better than grids in 3D[9].

The grid-based methods' calculation workload grows exponentially with dimension as K^d , becoming impractical for computers for $d>5$ (for instance, $d=10$, which demands 10^{10} grid points), whereas the computational requirement of Monte Carlo methods grows only linearly with N [10].

Conclusion

During the simple random walk experiment, the simulations produce results that confirm that the average distance and mean squared distance align with the laws $\langle r^2 \rangle = N$ and $\langle r \rangle = C_d \cdot \sqrt{N}$ (at fixed step size, $\sigma_s = 1$) in two dimensions. In two dimensions, when the

steps are distributed uniformly and randomly selected instead of fixed, the average distance still aligns with the law of $\langle r^2 \rangle = N \cdot \sigma_s^2$ and $\langle r \rangle = C_d \cdot \sigma_s \cdot \sqrt{N}$. With fixed steps under higher dimensions, the average distance decreases, and the spread decreases, while $\langle r^2 \rangle = N$ remains unchanged. The return-to-origin probabilities of different dimensions are also consistent with Pólya's recurrence theorem. For Monte Carlo, simulated 2D circle area and high-dimensional sphere volumes match theoretical volumes and error patterns. It is effective and accurate when applied to higher dimensions d . Grid methods lead to invalid results when applied to higher dimensions.

Reference

- [1] J. M. Hammersley, "The area enclosed by Pólya's walk," Mathematical Proceedings of the Cambridge Philosophical Society, vol. 52, no. 1, pp. 78–87, Jan. 1956, doi: 10.1017/S0305004100031005.
- [2] "Proof of $R = s\sqrt{N}$ | IOPSpark." Accessed: Mar. 24, 2026. [Online]. Available: <https://spark.iop.org/proof-r-sn>
- [3] K. J. B. Ghosh, S. Kais, and D. Herschbach, "Dimensional Interpolation for Random Walk," J. Phys. Chem. A, vol. 125, no. 34, pp. 7581–7587, Sep. 2021, doi: 10.1021/acs.jpca.1c05551.
- [4] "Return Probability of Quantum and Correlated Random Walks." Accessed: Mar. 24, 2026. [Online]. Available: <https://www.mdpi.com/1099-4300/24/5/584/xml>
- [5] R. Bakx, R. H. P. Kleiss, and F. Versteegen, "First- and second-order error estimates in Monte Carlo integration," Computer Physics Communications, vol. 208, pp. 29–34, Nov. 2016, doi: 10.1016/j.cpc.2016.07.021.
- [6] "Volume of a Ball in N Dimensions – Math Fun Facts." Accessed: Mar. 24, 2026. [Online]. Available: <https://math.hmc.edu/funfacts/volume-of-a-ball-in-n-dimensions/>
- [7] "1.1: Statistical Properties of Random Walks," Physics LibreTexts. Accessed: Mar. 24, 2026. [Online]. Available: [https://phys.libretexts.org/Bookshelves/Thermodynamics_and_Statistical_Mechanics/Book%3A_Thermodynamics_and_Statistical_Mechanics_\(Arovas\)/01%3A_Fundamentals_of_Probability/1.01%3A_Statistical_Properties_of_Random_Walks](https://phys.libretexts.org/Bookshelves/Thermodynamics_and_Statistical_Mechanics/Book%3A_Thermodynamics_and_Statistical_Mechanics_(Arovas)/01%3A_Fundamentals_of_Probability/1.01%3A_Statistical_Properties_of_Random_Walks)
- [8] T. Mitra, T. Hossain, S. Banerjee, and M. K. Hassan, "Similarity and self-similarity in random walk with fixed, random and shrinking steps," Chaos, Solitons & Fractals, vol. 145, p. 110790, Apr. 2021, doi: 10.1016/j.chaos.2021.110790.
- [9] "Volume of Unit Hypersphere - ProofWiki." Accessed: Mar. 24, 2026. [Online]. Available: https://proofwiki.org/wiki/Volume_of_Unit_Hypersphere
- [10] R. Schürer, "A comparison between (quasi-)Monte Carlo and cubature rule based methods for solving high-dimensional integration problems," Mathematics and Computers in Simulation, vol. 62, no. 3, pp. 509–517, Mar. 2003, doi: 10.1016/S0378-4754(02)00250-1.